Light

Network Fundamentals

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AQA GCSE Computer Science — **Fundamentals of Computer Networks (8525 Paper 2)**. This topic covers network definitions and types (PAN, LAN, WAN), wired and wireless connections, network protocols, and the TCP/IP model. You must be able to define, describe, explain, compare and discuss these concepts in exam questions.

Exam questions typically ask you to **define** what a network is, **describe** network types and their characteristics, **explain** the purpose of protocols, **compare** wired and wireless networks, and **discuss** the advantages and disadvantages of networking computers. Questions range from 1-mark definitions to 6-mark extended responses.

Why computer networks matter

A **computer network** is two or more computers connected together so they can communicate and share resources. Networks are fundamental to modern computing — from the internet that connects billions of devices worldwide, to the local network in a school that lets students access shared files and printers. Understanding networks helps you appreciate how data travels between devices, why different network types exist for different purposes, and how protocols ensure reliable communication. This knowledge is essential for understanding modern

digital infrastructure and the security considerations that come with connected systems.

- **PAN:** Personal Area Network — connects devices within a few metres (Bluetooth)
- **LAN:** Local Area Network — connects devices within a building or site
- **WAN:** Wide Area Network — connects devices across large geographical areas (the internet is the largest WAN)

What you must know

This checklist covers the key exam-visible concepts from the AQA specification. You should be able to explain each point and demonstrate each skill.

- **Define** what a computer network is.
- **Discuss** the advantages and disadvantages of computer networks.
- **Describe** the main types of computer network: Personal Area Network (PAN), Local Area Network (LAN), and Wide Area Network (WAN).
- **Understand** that networks can be wired or wireless.
- **Discuss** the advantages and disadvantages of wireless networks as opposed to wired networks.
- **Define** the term network protocol.
- **Explain** the purpose of common network protocols: TCP, IP, HTTP, HTTPS, SMTP, IMAP.
- **Describe** the 4-layer TCP/IP model and the function of each layer.
- **Understand** which protocols operate at which layer of the TCP/IP model.

> **Specification note.** AQA 8525 assesses computer networks on Paper 2 (Computing concepts). For PANs, only Bluetooth needs to be considered. You should know that wired networks can use different types of cable such as fibre and copper, and when each would be appropriate. Network security methods (authentication, encryption, firewall, MAC address filtering) are covered on this page but link closely to the cyber security topic.

Toolkit

Network types at a glance

Network type	Range	Ownership	Example
PAN	A few metres	Individual	Bluetooth headphones to phone
LAN	Building or site	Single person/organisation	School computer network
WAN	Large geographical area	Collective/distributed	The internet

Key protocols and their purposes

Protocol	Full name	Purpose	TCP/IP layer
TCP	Transmission Control Protocol	Sets up communication; ensures reliable data delivery	Transport
IP	Internet Protocol	Addresses and routes packets across networks	Internet
HTTP	Hypertext Transfer Protocol	Transfers web pages between browsers and servers	Application
HTTPS	Hypertext Transfer Protocol Secure	Encrypted version of HTTP for secure web transactions	Application
SMTP	Simple Mail Transfer Protocol	Sends emails between servers	Application
IMAP	Internet Message Access Protocol	Retrieves and stores emails on a server	Application

The 4-layer TCP/IP model

Layer	Name	Function	Protocols
4	Application	Where network applications operate (web browsers, email programs)	HTTP, HTTPS, SMTP, IMAP
3	Transport	Sets up communication between hosts; agrees packet size	TCP
2	Internet	Addresses and packages data; routes packets across network	IP
1	Link	Network hardware (NIC) and OS device drivers	—

Wired vs wireless comparison

Factor	Wired	Wireless
Speed	Generally faster and more consistent	Can be slower; affected by interference

Factor	Wired	Wireless
Security	More secure (physical access needed)	Less secure (signals can be intercepted)
Mobility	Devices must stay connected to cables	Devices can move freely within range
Installation	More complex; requires cable routing	Easier; no physical cables needed
Cost	Cabling can be expensive	Lower installation costs
Reliability	More reliable; fewer interference issues	Can suffer from interference and dropouts

Network security methods

Method	Description
Authentication	Verifying the identity of a user or device (e.g., username and password)
Encryption	Scrambling data so it cannot be read if intercepted
Firewall	Monitors incoming/outgoing traffic and blocks/allows based on security rules
MAC address filtering	Allows or blocks devices based on their unique physical hardware address

1. Network types and their characteristics

A **computer network** is two or more computers (or devices) connected together so that they can communicate with each other and share resources. Networks allow users to share files, access the internet, use shared printers, and communicate through email or messaging systems.

1.1 Defining a computer network

The AQA mark scheme requires a precise definition. A computer network is:

Definition: Two or more computers/devices connected together to communicate and share resources.

The key elements of this definition are: multiple devices, a connection between them, and the ability to communicate or share resources. Without all three elements, you do

not have a network.

1.2 Advantages of computer networks

Networks provide many benefits for organisations and individuals:

- **File sharing:** Users can access shared files and folders from any networked computer
- **Resource sharing:** Peripheral devices like printers and scanners can be shared, reducing costs
- **Internet access:** A single internet connection can be shared across all networked devices
- **Central backup:** Files can be backed up centrally, ensuring data is protected
- **Central management:** Software and settings can be managed and updated from one location
- **Communication:** Users can communicate via email, messaging, and video conferencing
- **User accounts:** Users can log in to any networked computer and access their files and settings
- **Cost savings:** Fewer peripherals needed; cheaper software licensing for networks

1.3 Disadvantages of computer networks

Networks also introduce challenges and risks:

- **Security risks:** Networks are vulnerable to hacking, malware, and unauthorised access
- **Server dependency:** If the server fails, users may lose access to files and services
- **Malware spread:** Viruses and malware can spread quickly across networked computers
- **Setup costs:** Initial costs for hardware, cabling, and configuration can be high
- **Maintenance:** Networks require ongoing management and may need a dedicated administrator
- **Performance:** Network speed can slow down as more users access resources simultaneously
- **Hardware dependence:** Failure of network hardware (switches, routers) affects all users

1.4 Personal Area Network (PAN)

A **Personal Area Network (PAN)** is the smallest type of network, connecting devices within a very short range — typically a few metres. For AQA, only Bluetooth needs to be considered as the technology for PANs.

Key characteristics of a PAN:

- Range: typically up to 10 metres
- Technology: Bluetooth (for AQA)
- Ownership: controlled by an individual
- Purpose: connecting personal devices

Examples of PAN connections include:

- Wireless headphones connected to a smartphone via Bluetooth
- A smartwatch syncing with a phone
- A wireless keyboard connected to a tablet
- A fitness tracker sending data to a phone

1.5 Local Area Network (LAN)

A **Local Area Network (LAN)** covers a relatively small geographical area, such as a single building, school, office, or home. LANs are typically owned and controlled by a single person or organisation.

Key characteristics of a LAN:

- Range: a building or small site (e.g., school campus)
- Ownership: usually owned and managed by a single organisation
- Speed: typically high-speed connections
- Cost: the organisation pays for all hardware and maintenance

Examples of LANs include:

- A school network connecting computers in classrooms, the library, and offices
- A home network connecting a router to computers, phones, and smart devices
- An office network connecting workstations, printers, and servers
- A small business network in a shop or restaurant

LANs can use wired connections (Ethernet cables), wireless connections (Wi-Fi), or a combination of both. The organisation that owns the LAN is responsible for its security, maintenance, and management.

1.6 Wide Area Network (WAN)

A **Wide Area Network (WAN)** covers a large geographical area, potentially spanning cities, countries, or even the entire world. The internet is the largest and most well-known example of a WAN.

Key characteristics of a WAN:

- Range: large geographical areas (cities, countries, worldwide)
- Ownership: often under collective or distributed ownership
- Infrastructure: uses public communication links (telephone lines, satellites, undersea cables)
- Speed: can vary; often slower than LANs due to distance

The internet is the biggest example of a WAN. It connects billions of devices worldwide through a complex infrastructure of cables, satellites, and data centres. No single organisation owns the internet — it is maintained collectively by many different organisations, governments, and service providers.

Other examples of WANs include:

- A bank's network connecting branches across a country
- A multinational company's network linking offices worldwide
- Government networks connecting departments in different locations

1.7 Comparing PAN, LAN, and WAN

Understanding the differences between network types is essential for exam questions that ask you to compare or identify appropriate network types for given scenarios.

Feature	PAN	LAN	WAN
Geographical range	Few metres	Building/site	Cities/countries/worldwide
Typical ownership	Individual	Single organisation	Collective/distributed

Feature	PAN	LAN	WAN
Example technology	Bluetooth	Ethernet, Wi-Fi	Internet infrastructure
Speed	Moderate	High	Variable
Security control	User controlled	Organisation controlled	Varied/complex

Worked Example 1 – Defining a computer network

Question: Define what is meant by a computer network. [2 marks]

Model answer:
Two or more computers/devices **(1)** connected together to communicate and/or share resources **(1)**.

Exam tip: Both parts are needed for full marks. "Connected computers" alone only scores 1 mark — you must mention communication or resource sharing.

Worked Example 2 – Identifying network types

Question: State the type of network (PAN, LAN, or WAN) that best describes each of the following:
(a) A student's phone connected to their Bluetooth earphones. [1 mark]
(b) The computer network in a school building. [1 mark]
(c) The internet. [1 mark]

Model answer:
(a) **PAN** — Personal Area Network (Bluetooth connection, few metres range)
(b) **LAN** — Local Area Network (single building, owned by one organisation)
(c) **WAN** — Wide Area Network (worldwide coverage, distributed ownership)

Exam tip: Look for clues: Bluetooth = PAN; single building/site = LAN; large area/internet = WAN.

Worked Example 3 – Advantages of networks

Question: A small business is considering connecting its five computers into a network. State two advantages of doing this. [2 marks]

Model answer:
1. Files can be shared between computers/employees can access shared documents **(1)**
2. Peripheral devices such as printers can be shared, reducing costs **(1)**

Exam tip: Make sure your two points are clearly different. Other valid answers include: central backup, internet sharing, central software management, user accounts accessible from any computer.

Worked Example 4 — Disadvantages of networks

Question: State two disadvantages of setting up a computer network. [2 marks]

Model answer:
1. Increased security risks from hacking or malware spreading across the network **(1)**
2. If the server fails, all users may lose access to files and services **(1)**

Exam tip: Other valid answers include: setup costs, need for network administrator, malware can spread more easily, network hardware failure affects all users.

Extended response practice. The AQA exam includes 6-mark extended response questions on networks. Here is an example with a model banded answer.

Extended Response — Network advantages and disadvantages in context

Question: A school currently has standalone computers in each classroom. The head teacher is considering installing a network to connect all the computers. Discuss the advantages and disadvantages to the school of installing a network instead of using standalone computers. [6 marks]

Model answer (Level 3, 5—6 marks):
Installing a network would provide several advantages for the school. Students and staff could store their files centrally on a server, meaning they could access their work from any computer in the school rather than being tied to one machine. This would be particularly useful for students who use different classrooms for different lessons.
The school could also share peripheral devices such as printers between multiple computers, reducing costs as fewer printers would be needed.
Software could be installed and updated centrally by the IT technician, saving

time compared to updating each standalone computer individually. Staff could also access school resources from home if the network is configured appropriately.

However, there are disadvantages to consider. If the server fails, all users would lose access to their files and shared resources, potentially disrupting lessons across the whole school. The network would also introduce security risks — malware could spread more easily between connected computers, and the school would need to protect against hacking attempts to access student and staff data.

The school would need to invest in network hardware such as switches and cabling, and may need to employ or train a network administrator to maintain the system. Network speeds could also slow down if many students access resources simultaneously.

Overall, while the benefits of file sharing, central management and reduced peripheral costs are significant for a school environment, these must be weighed against the security risks, setup costs, and dependency on the server and network hardware working reliably.

Exam tip (6-mark questions): To reach Level 3 (5—6 marks), you must discuss BOTH advantages AND disadvantages, link your points to the specific context (school/staff/students), and structure your response clearly. Generic points without context will only reach Level 1—2.

2. Wired and wireless networks

Networks can be connected using physical cables (wired) or radio signals (wireless). Each approach has advantages and disadvantages, and many modern networks use a combination of both.

2.1 Wired networks

Wired networks use physical cables to connect devices. The two main types of cable used are **copper** (typically Ethernet cables) and **fibre optic** cables.

Copper cables (Ethernet)

Copper Ethernet cables are the most common type of wired connection in LANs. They transmit data as electrical signals through copper wires.

- **Speed:** High speeds suitable for most network tasks
- **Range:** Typically up to 100 metres per cable run
- **Cost:** Relatively inexpensive
- **Installation:** Easy to terminate and install
- **Interference:** Can be affected by electromagnetic interference

Copper cables are appropriate for most office and school environments where distances are short and very high speeds are not essential.

Fibre optic cables

Fibre optic cables transmit data as pulses of light through thin glass or plastic fibres. They offer significant advantages over copper for certain applications.

- **Speed:** Much faster than copper

- **Range:** Can transmit over much longer distances (kilometres) without signal loss
- **Interference:** Not affected by electromagnetic interference
- **Security:** More difficult to tap or intercept
- **Cost:** More expensive than copper
- **Installation:** Requires specialist equipment and skills to install

Fibre optic cables are appropriate for high-speed backbone connections, connections between buildings, and situations where long distances or high bandwidth are required.

When to use copper vs fibre

Use copper when...	Use fibre when...
Distances are short (under 100m)	Long distances are involved
Cost is a primary concern	Very high speeds are required
Standard office/classroom connections	Connecting buildings or backbone infrastructure
Existing copper infrastructure is in place	Electromagnetic interference is a problem

2.2 Wireless networks

Wireless networks use radio signals to transmit data between devices. Wi-Fi is the most common wireless technology for LANs, while Bluetooth is used for PANs.

- **Mobility:** Devices can move freely within the signal range
- **Installation:** No cables to route through walls or ceilings
- **Flexibility:** Easy to add new devices to the network
- **Interference:** Can be affected by walls, other devices, and competing signals
- **Security:** Signals can potentially be intercepted; encryption is essential
- **Speed:** Generally slower than wired connections

2.3 Advantages of wireless networks over wired

Wireless networks offer several benefits compared to wired alternatives:

- **Mobility:** Users can move around while staying connected to the network
- **Easier installation:** No need to install cables throughout a building
- **Lower installation cost:** No expense for cabling and cable routing
- **Flexibility:** Easy to add new devices without running additional cables
- **No physical constraints:** Devices are not tied to specific locations
- **Easier to scale:** Expanding the network doesn't require new cable runs
- **Guests and visitors:** Can easily provide temporary network access

2.4 Disadvantages of wireless networks compared to wired

Wireless networks also have significant drawbacks:

- **Security risks:** Signals can be intercepted; requires strong encryption
- **Slower speeds:** Generally slower than wired connections, especially with many users
- **Interference:** Walls, other electronic devices, and neighbouring networks can cause interference
- **Range limitations:** Signal strength decreases with distance

- **Less reliable:** Connection dropouts and speed fluctuations are more common
- **Bandwidth sharing:** All wireless devices share the available bandwidth

2.5 Network security methods

Networks, especially wireless networks, require security measures to protect against unauthorised access and data interception. AQA requires knowledge of four key security methods:

Authentication

Authentication is the process of verifying the identity of a user or device. The most common form is username and password authentication, but it can also include:

- Two-factor authentication (password plus code sent to phone)
- Biometric authentication (fingerprint, face recognition)
- Smart cards or security tokens

Authentication ensures that only authorised users can access network resources.

Encryption

Encryption scrambles data so that it cannot be read if intercepted. The original data (plaintext) is converted into an unreadable form (ciphertext) using an encryption algorithm and key. Only someone with the correct key can decrypt the data back to its original form.

Encryption protects data during transmission, so even if someone intercepts the network traffic, they cannot understand the content. HTTPS uses encryption to secure web traffic, and WPA2/WPA3 encrypt wireless network communications.

Firewall

A **firewall** is a network security device (hardware or software) that monitors incoming and outgoing network traffic. It decides whether to allow or block specific traffic based on a defined set of security rules.

Firewalls can:

- Block access to certain websites or services
- Prevent unauthorised external connections to the network
- Block suspicious traffic patterns that might indicate an attack
- Log network activity for security monitoring

MAC address filtering

Every network interface card (NIC) has a unique physical address called a **MAC (Media Access Control) address**. MAC address filtering allows network administrators to:

- Create a list of approved MAC addresses that are allowed to connect
- Block specific MAC addresses from accessing the network

This provides an additional layer of security by controlling which physical devices can connect, based on their hardware address embedded in the network adapter.

> **Security in combination.** These security methods work together to provide greater protection. For example, a school network might use authentication (username/password), encryption (WPA2 for Wi-Fi), a firewall (blocking inappropriate websites), and MAC filtering (only allowing registered school devices).

> #### Worked Example 5 — Wired vs wireless advantages
>
> **Question:** State two advantages of using a wireless network instead of a wired network in a school. [2 marks]

Model answer:
1. Students and staff can move around the building while staying connected to the network **(1)**
2. No need to install cables throughout the building, reducing installation costs **(1)**

Exam tip: Link your answers to the context where possible. For a school: mobility for students, easy connection of devices brought from home (BYOD), flexibility to rearrange rooms.

Worked Example 6 — Wired vs wireless disadvantages

Question: State two disadvantages of using a wireless network compared to a wired network. [2 marks]

Model answer:
1. Wireless signals can be intercepted, making it less secure than wired connections **(1)**
2. Wireless networks are generally slower than wired networks, especially with many users **(1)**

Exam tip: Other valid answers: interference from walls/other devices, less reliable/more dropouts, range limitations.

Worked Example 7 — Cable types

Question: A company needs to connect two office buildings that are 500 metres apart. Explain why fibre optic cable would be more suitable than copper Ethernet cable for this connection. [2 marks]

Model answer:
Fibre optic can transmit over longer distances without signal loss **(1)**, while copper Ethernet is limited to approximately 100 metres per cable run **(1)**.

Exam tip: Other valid points: fibre is faster, fibre is not affected by electromagnetic interference. Always relate back to the scenario — here, distance is the key factor.

Worked Example 8 — Network security methods

Question: Explain how encryption can make the transmission of data across a network more secure. [2 marks]

Model answer:
Encryption scrambles the data before transmission / converts plaintext to ciphertext **(1)**, so if the data is intercepted it cannot be read/understood by unauthorised users **(1)**.

Exam tip: The AQA mark scheme requires both parts: what encryption does (scrambles data) AND why that makes it secure (cannot be read if intercepted). Just saying "encrypts the data" is not enough.

3. Network protocols and the TCP/IP model

For devices on a network to communicate successfully, they must follow agreed rules and standards. These rules are called **protocols**. Without protocols, devices from different manufacturers using different software would not be able to understand each other.

3.1 What is a network protocol?

A **network protocol** is a set of rules that govern how data is transmitted and received across a network. Protocols define:

- The format of data packets
- How devices identify themselves
- How errors are detected and handled
- How communication sessions are started and ended
- The sequence of operations in data transmission

Definition: A network protocol is a set of rules that govern/control how data is transmitted across a network.

3.2 TCP (Transmission Control Protocol)

TCP is responsible for reliable data transmission between devices. It operates at the Transport layer of the TCP/IP model.

TCP performs several important functions:

- **Connection setup:** Establishes a connection between sender and receiver before data transfer begins
- **Packet management:** Breaks data into packets and reassembles them at the destination
- **Error checking:** Ensures all packets arrive correctly and requests retransmission of any lost or corrupted packets
- **Flow control:** Manages the rate of data transmission to prevent overwhelming the receiver
- **Ordered delivery:** Ensures packets are reassembled in the correct order

TCP is used when reliable delivery is essential, such as for web pages, emails, and file transfers.

3.3 IP (Internet Protocol)

IP is responsible for addressing and routing packets across networks. It operates at the Internet layer of the TCP/IP model.

IP performs these key functions:

- **Addressing:** Assigns unique IP addresses to identify devices on a network
- **Packaging:** Packages data into packets with header information including source and destination addresses
- **Routing:** Determines the path packets take across the network to reach their destination

Every device on the internet has an IP address that identifies it uniquely on the network. IP addresses allow data to be sent to the correct destination.

3.4 HTTP (Hypertext Transfer Protocol)

HTTP is the protocol used for transferring web pages between web browsers and web servers. It operates at the Application layer.

- When you type a web address into a browser, an HTTP request is sent to the web server
- The server responds with the requested web page content
- HTTP is a "request-response" protocol — the client requests, the server responds

HTTP transmits data in plain text, which means it is not secure. Anyone intercepting the traffic can read the content.

3.5 HTTPS (Hypertext Transfer Protocol Secure)

HTTPS is an encrypted version of HTTP that provides secure web communication. It also operates at the Application layer.

- HTTPS encrypts all data transmitted between browser and server
- Used for sensitive transactions: online banking, shopping, login pages
- Websites using HTTPS display a padlock icon in the browser address bar
- Protects against eavesdropping and man-in-the-middle attacks

The key difference from HTTP is that HTTPS adds encryption, making the transmitted data unreadable to anyone who intercepts it.

3.6 SMTP (Simple Mail Transfer Protocol)

SMTP is the protocol used to send emails between mail servers. It operates at the Application layer.

- When you send an email, SMTP transfers it from your mail server to the recipient's mail server
- SMTP is only used for sending emails, not receiving them
- Email clients use SMTP to send outgoing mail to the mail server

3.7 IMAP (Internet Message Access Protocol)

IMAP is the protocol used to retrieve and manage emails from a mail server. It operates at the Application layer.

- IMAP stores emails on the server rather than downloading them to the device
- Users can view and manipulate emails as if they were stored locally
- Emails are synchronised across multiple devices (phone, tablet, computer)
- Changes made on one device (reading, deleting, moving) are reflected on all devices

IMAP's server-based storage is well suited to modern multi-device usage, allowing users to access the same emails from any device.

SMTP vs IMAP — understanding the difference

SMTP: Sends emails between servers

IMAP: Retrieves/stores emails on server; allows viewing on multiple devices

Think of it this way: SMTP is for sending (S for Sending), IMAP is for accessing/managing your inbox.

3.8 The 4-layer TCP/IP model

The **TCP/IP model** describes how data is transmitted across networks. It divides network communication into four layers, each with specific responsibilities. Data passes down through the layers when being sent, and up through the layers when being received.

Layer 4: Application layer

This is where network applications operate, such as web browsers and email programs. The application layer provides the interface between the user's software and the network.

- **Function:** Provides network services directly to user applications
- **Protocols:** HTTP, HTTPS, SMTP, IMAP
- **Examples:** Web browsing, sending/receiving email

Layer 3: Transport layer

This layer sets up communication between the two hosts (sending and receiving devices). The hosts agree on settings such as the size of packets and how to handle errors.

- **Function:** Establishes communication; manages reliable data delivery
- **Protocol:** TCP
- **Responsibilities:** Breaking data into packets, error checking, flow control, packet reassembly

Layer 2: Internet layer

This layer handles addressing and routing. It packages data for transmission and determines the route packets take across the network to reach their destination.

- **Function:** Addresses and packages data; routes packets across the network
- **Protocol:** IP
- **Responsibilities:** IP addressing, packet routing, fragmentation

Layer 1: Link layer

This is where the network hardware operates. It includes the physical network interface card (NIC) and the operating system device drivers that control it.

- **Function:** Handles physical transmission of data
- **Components:** Network Interface Card (NIC), device drivers
- **Responsibilities:** Physical addressing (MAC addresses), converting data to signals for transmission

Remembering the TCP/IP layers

From top to bottom: **A**pplication → **T**ransport → **I**nternet → **L**ink

Memory aid: "**A** **T**iny **I**nsect **L**anded" or simply remember the protocols at each layer.

Layer	Protocols	Key word to remember
Application	HTTP, HTTPS, SMTP, IMAP	User applications
Transport	TCP	Reliable delivery
Internet	IP	Addressing and routing

Layer	Protocols	Key word to remember
Link	—	Hardware (NIC)

3.9 How the layers work together

When you visit a website, all four layers work together:

1. **Application layer:** Your browser uses HTTP/HTTPS to request the web page
2. **Transport layer:** TCP breaks the request into packets and manages reliable delivery
3. **Internet layer:** IP adds addressing information and routes packets to the web server
4. **Link layer:** The NIC converts data to electrical or optical signals for transmission

At the receiving end, the process reverses — data moves up through the layers until the web page is displayed in your browser.

Worked Example 9 – Defining a protocol

Question: Define the term network protocol. [1 mark]

Model answer:
A set of rules that govern/control how data is transmitted across a network **(1)**.

Exam tip: The key phrase is "set of rules" — this must be included. Adding "for data transmission" or "across a network" shows you understand the purpose.

Worked Example 10 – SMTP and IMAP

Question: Describe the purpose of SMTP and IMAP. [2 marks]

Model answer:
SMTP: Sends emails between servers **(1)**
IMAP: Stores emails on a server / retrieves emails / allows viewing emails on multiple devices **(1)**

Exam tip: SMTP = Sending. IMAP = Inbox Access/Management. Don't confuse them — this is a common exam error.

Worked Example 11 – TCP/IP layers

Question: State the four layers of the TCP/IP model in order, starting from the layer closest to the user. [2 marks]

Model answer:

1. Application layer
2. Transport layer
3. Internet layer
4. Link layer

(1 mark for any two correct in sequence, 2 marks for all four correct in sequence)

Exam tip: Check whether the question asks for top-to-bottom (Application first) or bottom-to-top (Link first). Read carefully!

Worked Example 12 – Protocol purposes

Question: Explain why HTTPS is more suitable than HTTP for online banking. [2 marks]

Model answer:
HTTPS encrypts the data being transmitted **(1)**, so sensitive information such as passwords and account details cannot be read if intercepted **(1)**.

Exam tip: Always link the technical feature (encryption) to the benefit in context (protecting sensitive banking data). Both parts are needed for full marks.

Worked examples summary

This page contains 12 worked examples plus one extended response, covering all the key skills for computer networks. Use this table to find specific types of questions for revision.

Example	Topic	Question type	Marks
1	Defining a network	Define	2
2	Identifying PAN, LAN, WAN	State/Identify	3
3	Advantages of networks	State	2
4	Disadvantages of networks	State	2
Ext.	Network advantages/disadvantages in context	Discuss (extended response)	6
5	Wireless advantages	State	2
6	Wireless disadvantages	State	2

Example	Topic	Question type	Marks
7	Copper vs fibre cables	Explain	2
8	Encryption security	Explain	2
9	Defining protocol	Define	1
10	SMTP and IMAP	Describe	2
11	TCP/IP layers	State	2
12	HTTPS vs HTTP	Explain	2

Quick reference

Key definitions (mark-scheme style)

Term	Definition
Computer network	Two or more computers/devices connected together to communicate and share resources.
PAN	Personal Area Network — connects devices within a few metres using Bluetooth.
LAN	Local Area Network — covers a building or site; owned by a single organisation.
WAN	Wide Area Network — covers large geographical areas; often distributed ownership. The internet is the largest WAN.
Network protocol	A set of rules that govern how data is transmitted across a network.
TCP	Transmission Control Protocol — sets up communication and ensures reliable data delivery.
IP	Internet Protocol — addresses and routes packets across networks.
HTTP	Hypertext Transfer Protocol — transfers web pages between browsers and servers.
HTTPS	Hypertext Transfer Protocol Secure — encrypted version of HTTP for secure web transactions.
SMTP	Simple Mail Transfer Protocol — sends emails between servers.
IMAP	Internet Message Access Protocol — retrieves and stores emails on a server; allows multi-device access.

Term	Definition
Authentication	Verifying the identity of a user or device (e.g., username and password).
Encryption	Scrambling data so it cannot be read if intercepted.
Firewall	Monitors network traffic and blocks/allows based on security rules.
MAC address filtering	Allowing or blocking devices based on their unique physical hardware address.

TCP/IP model summary

Layer	Name	Function	Protocols
4 (top)	Application	Network applications operate here (browsers, email)	HTTP, HTTPS, SMTP, IMAP
3	Transport	Sets up communication; reliable delivery; packet management	TCP
2	Internet	Addressing and packaging; routing packets	IP
1 (bottom)	Link	Network hardware (NIC); device drivers	—

Network advantages and disadvantages

Advantages

- File sharing and central storage
- Shared peripherals (printers)
- Shared internet connection
- Central backup
- Central software management
- User accounts on any computer
- Communication (email, messaging)

Disadvantages

- Security risks (hacking, malware)
- Server failure affects all users
- Malware spreads more easily
- Setup and hardware costs
- Need for network administrator
- Performance degrades with usage

- Dependence on network hardware

Wired vs wireless summary

Wired advantages

- Faster, more consistent speeds
- More secure (physical access needed)
- More reliable (less interference)
- Not affected by walls/distance

Wireless advantages

- Mobility — users can move freely
- Easier/cheaper installation
- Easy to add new devices
- Flexible device positioning

Typical pitfalls

Pitfall 1 — Incomplete network definition.
Wrong: "A network is computers connected together."
Right: Two or more devices connected together to communicate and/or share resources. Include the purpose!

Pitfall 2 — Confusing LAN and WAN ownership.
Wrong: "WANs are owned by one company."
Right: LANs are owned by a single organisation. WANs often have collective/distributed ownership.

Pitfall 3 — Saying wireless is "more secure".
Wrong: "Wireless is more secure because there are no cables to tap."
Right: Wired is generally more secure because physical access is needed. Wireless signals can be intercepted.

Pitfall 4 — Confusing SMTP and IMAP.
Wrong: "SMTP is for receiving emails."
Right: SMTP sends emails. IMAP retrieves/stores emails on the server.

Pitfall 5 — Vague protocol definitions.
Wrong: "HTTP is for the internet."

Right: HTTP transfers web pages between browsers and web servers. Be specific about what each protocol does.

Pitfall 6 — Forgetting encryption's two parts.
Wrong: "Encryption encrypts the data."
Right: Encryption scrambles data (1) so it cannot be read if intercepted (2). Both parts are needed for full marks.

Pitfall 7 — Wrong TCP/IP layer order.
Wrong: Listing layers in random order.
Right: Top to bottom: Application → Transport → Internet → Link. Check which direction the question asks for.

Pitfall 8 — Forgetting PAN uses Bluetooth.
Wrong: "PANs use Wi-Fi."
Right: For AQA, PANs use Bluetooth. Wi-Fi is typically used for LANs.

Pitfall 9 — Not linking answers to context.
Wrong: Generic advantages that don't mention the scenario.
Right: If asked about a school network, mention students, staff, classrooms, shared resources for education.

Pitfall 10 — Confusing HTTP and HTTPS.
Wrong: "HTTP is secure."
Right: HTTP is not encrypted. HTTPS is the secure (encrypted) version. The 'S' stands for Secure.

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